5. Red thread · Method [ALG] Program: choose a number x ; Add 3 ; multiply by 2 ; subtract 6. Calculate the result for $x = 5$, then write an expression in x What do we notice?

Rep. For 5: 10. Expression: 2 $(x+3)-6=2x$: we always get twice the number chosen.

Link A09: Literal writing proves what digital essays suggest.

Form 12 Mar. 17 Nov. 2026 Intersection of two lines; Python function of alignment

1. Warming · Flash [EQ] Solve 2 $x+1=-x+7$.

Rep. $3x=6$, so $x=2$.

2. Heart · Method [GEO] Contact details of the intersection of $y=2x+1$ and $y=-x+7$.

Rep. $x=2$ then $y=2\times 2+1=5$: $(2;5)$.

Control in the second equation: $-2+7=5$.

3. Heart + · Code reading [ALG] What Returned line(0.0,1,2,2.4) and lines(0.0,1,2,2.5) ?

```
def lines(xa, ya, xb, yb, xc, yc):
    return (xb-xa)*(yc-ya) - (yb-ya)*(xc-xa) = 0
```

Rep. True then False : determinants $1\times 4-2\times 2=0$ and $1\times 5-2\times 2=1\neq 0$.

4. Back · Develop [DEV] Develop and reduce 3 $(x-2)$.

Rep. $3x-6$.

5. Red thread · Prove [LIT] Show that the sum of three consecutive integers is a multiple of 3.

Rep. With $n-1$, n, $n+1$: sum $=3n$, multiple of 3.

Selecting the term of the medium simplifies the calculation.

Actual numbers and intervals

Form 13 Thu 19 Nov 2026 Sets of numbers: $\mathbb{N},\mathbb{Z},\mathbb{D},\mathbb{Q},\mathbb{R}$

1. Warming · Flash [NUM] Give decimal writing of $\frac{3}{8}$ and an approximate value of $5\frac{}{6}$ à 10^{-3} Close.

Rep. 0,375 ; 0,833.

2. Heart · Method [NUM] Give the smallest set ($\mathbb{N},\mathbb{Z},\mathbb{D},\mathbb{Q},\mathbb{R}$) containing each of these numbers : -4 ; 0,25 ; $\frac{1}{3}$; $\sqrt{2}$; $\frac{6}{3}$.

Rep. $-4\in\mathbb{Z}$; $0{,}25\in\mathbb{D}$; $\frac{1}{3}\in\mathbb{Q}$; $\sqrt{2}\in\mathbb{R}$; $\frac{6}{3}=2\in\mathbb{N}$.

The smallest set is the key: $\frac{6}{3}$ is a natural integer.

3. Heart + · True/false [NUM] True or false? Justify. $\frac{1}{3}$ is a decimal number.

Rep. False. If there existed a and n integer with $\frac{1}{3}=\frac{a}{10^n}$, then $10^n=3a$ would be divisible by 3 : impossible (the sum of figures of 10^n is equal to 1).

4. Back · Flash [SUB] Calculate $(x+1)(x-3)$ for $x=2$.

Rep. -3.

5. Red thread · Flash [SOL] How many faces, peaks and edges have a triangular-based straight prism?

Rep. 5 sides (2 triangles, 3 rectangles), 6 tops, 9 edges.

Form 14 Mon. 23 Nov. 2026 Decimal and rounded frameworks

1. Warming · Flash [NUM] Round 3,14159 to 10^{-2} next to 10^{-3} Close.

Rep. 3,14 then 3,142.

The following figure decides: $1<5$ *puis* $5\geqslant 5$.

2. Heart · Method [NUM] Framework $\sqrt{2}$ by two decimals of amplitude 10^{-2}.

Rep. $1{,}41^2=1{,}9881<2<2{,}0164=1{,}42^2$: $1{,}41<\sqrt{2}<1{,}42$.

3. Heart + · Code reading [ALG] What is the final value of `k` ? What can be deduced on $\sqrt{2}$?

```
k = 10
while (k/10)**2 < 2:
    k = k + 1
```

Rep. $k = 15$: $1{,}4^2 = 1{,}96 < 2$ and $1.5^2 = 2{,}25 \geqslant 2$, therefore $1.4 < \sqrt{2} \leqslant 1{,}5$.

4. Back · Flash [QP] 5 kg of rice cost 12,500 Ar. How much is 8 kg?

Rep. $\frac{12\,500}{5} \times 8 = 20{,}000$ Ar.

5. Red thread · True/false [DEM] True or false? Justify. A quadrilateral whose diagonals even have length is a rectangle.

Rep. False. Counter-example: an isocele trapeze has diagonals of the same length without being a rectangle.

Form 15 Mar. 24 Nov. 2026 Irrational numbers: $\frac{1}{3}$ is not decimal, $\sqrt{2}$ is irrational

1. Warming · Flash [CAL] Write $\sqrt{12}$ and $\sqrt{50}$ in form $a\sqrt{b}$ with b integer as small as possible.

Rep. $\sqrt{12} = 2\sqrt{3}$; $\sqrt{50} = 5\sqrt{2}$.

2. Heart · Sort [NUM] Between 0.333 . . . ; $\sqrt{9}$; $\sqrt{2}$; π ; $\frac{22}{7}$ Which are rational?

Rep. $0{,}333\ldots = \frac{1}{3}$; $\sqrt{9} = 3$; $\frac{22}{7}$: rational. $\sqrt{2}$ and π are irrational.

$\frac{22}{7}$ is only a value close to π not equal to π.

3. Heart + · Reasoning [DEM] We assume. $\sqrt{2} = \frac{p}{q}$ with $\frac{p}{q}$ Indomitable, where $p^2 = 2q^2$. What to say about the gender parity p and then q What contradiction?

Rep. p^2 pair, so p pair, $p = 2k$; then 4 $k^2 = 2q^2$, $q^2 = 2k^2$: q pair. Contradiction with fraction I can't believe it.

4. Back · Table [STA] Values 1; 2; 3; 4; 5; 8; 4; 3. value 2 and the average.

Value	1	2	3	4
Count	5	8	4	3

Rep. Total 20; frequency $\frac{8}{20} = 40\%$; average 5 $\frac{+16+12+12}{20} = 2{,}25$.

5. Red thread · Flash [REP] Give the coordinates of the symmetrical of $A(3;-2)$ in relation to the axis of Ordered.

Rep. $(-3;-2)$.

Trap: change the sign of the order (symmetry to the axis of the abscisses).

Form 16 Thu. 26 Nov. 2026 Intervals $\mathbb{R}$: ratings and representation

1. Warming · Flash [CMP] Compare -3 and -5 then $-\frac{1}{2}$ and $-0{,}6$.

Rep. $-3 > -5$; $-0{,}5 > -0{,}6$.

For negatives, the smallest in absolute value is the largest.

2. Heart · Method [NUM] Write as an interval: $-2 \leqslant x < 5$; $x > 3$; $x \leqslant 0$.

Rep. $[-2;5[\,;\,]3;+\infty[\,;\,]-\infty;0]$.

The hook is turned inward if the bound is included; ∞ is always excluded.

3. Heart + · MCQ [NUM] The intersection $]-\infty;2[\,\cap\,[-1;4]$ is:

A. $]2;4]$ **B.** $]-\infty;4]$ **C.** $[-1;2[$ **D.** $[-1;2]$

Rep. C. $[-1;2[$

Intersection: values checking both conditions x < 2 *et* $-1 \leqslant x \leqslant 4$.

4. Back · Table [LEC] Travel time (min).